

September 9, 2026

Dora Matzke, PhD
Co-Editor-in-Chief
Joshua R. de Leeuw, PhD
Action Editor
and Editors of the Tutorial Collection
Behavior Research Methods

Re: Tutorial submission, “A Guided Search vision module for ACT-R: Implementation, tutorial, and validation against benchmark visual search data”

Dear Dr. Matzke, Dr. de Leeuw, and Editors of the Tutorial Collection:

On behalf of my coauthor, I am submitting our manuscript, “A Guided Search vision module for ACT-R: Implementation, tutorial, and validation against benchmark visual search data,” for consideration as a Tutorial in *Behavior Research Methods*. We are submitting the paper specifically for the journal's tutorial special issue announced through the “Call for Tutorial Papers” (Tutorial Collection).

The manuscript introduces gs-vision, an extension of the ACT-R 7.31 vision module that brings state-of-the-art visual-search cognitive mechanisms into a widely used cognitive architecture. It integrates feature-guided selection, visual acuity, capacity-limited identification, memory for rejected items, adaptive search termination, and eye movements while retaining ACT-R's existing visual buffers and ordinary requests. The paper gives researchers a step-by-step guide to installing and using the module, describing displays, issuing searches, inspecting diagnostics, running experiments from Python, and fitting parameters to behavioral data.

We believe the manuscript is a strong fit for both *Behavior Research Methods* and the Tutorial Collection. It provides a practical guide to an open software tool and a reusable workflow for a common research challenge: building process-level models of how people search visual displays. It also goes beyond software documentation by evaluating the module against reaction-time distributions and error rates from three benchmark visual-search tasks. Under a commonly used held-out-participant evaluation protocol, gs-vision provides the most accurate ACT-R vision-module account among the alternatives examined. The manuscript reports the model's limitations as well, including its miss-rate and eye-movement discrepancies, so readers can judge when the module is appropriate for their work.

The submission follows the journal's emphasis on transparency and practical reuse. The module, comparison models, fitting and evaluation code, run records, and generated artifacts are publicly available at <https://github.com/ar-zadeh/gs-vision>. The repository also documents the public human datasets used in the validation and provides commands for regenerating the reported analyses and tests.

This manuscript is original, is not under consideration by another journal, and has been approved by both authors. The authors declare no competing interests. The study analyzes previously published, de-identified data and did not involve new data collection from human participants.

Thank you for considering our manuscript for the tutorial special issue and the *Behavior Research Methods* Tutorial Collection. We believe it will be useful to researchers in cognitive modeling, visual attention, experimental psychology, and human–computer interaction.

Sincerely,

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On behalf of Amirreza Bagherzadeh and Frank E. Ritter